

Dry Run Exercises

Unit 6 While loops

Complete the dry run table for each fragment of code given in the questions below. **Give** the value(s) printed out when each is executed.

Follow the module dry run style guide in your answers.

1.

```
int count = 0;                                // L1

while(count < 3)                               // L2
{
    count = count + 1;                         // L3
    System.out.println(count);                 // L4
}
```

Complete the dry run table (you may not need all rows):

LINE	count	count < 3 ?	printed

Write EXACTLY what is printed here

2.

```
String s = "ABA";           // L1

while(s.equals("ABA"))      // L2
{
    s = s + s;              // L3
}

System.out.println(s);      // L4
```

Complete the dry run table (you may not need all rows):

LINE	s	s.equals("ABA") ?	printed

Write EXACTLY what is printed here

3.

```
int t = 3;                // L1

while(t<=8)               // L2
{
    t = t + 2;            // L3
}

System.out.println(t);    // L4
```

Complete the dry run table (you may not need all rows):

LINE	t	t <= 8 ?	printed

Write EXACTLY what is printed here

```

4.
  int t = 3;                                // L1

  while(t<=8)                               // L2
  {
    t = t + 2;                              // L3
    System.out.println(t);                  // L4
  }

  System.out.println(t);                     // L5

```

Complete the dry run table (you may not need all rows):

LINE	t	t <= 8	printed

Write EXACTLY what is printed here

5.

```
int n = 0;                                // L1

while(n <= 3)                             // L2
{
    System.out.print(n);                  // L3
    n = n + 2;                           // L4
}
System.out.print(n);                      // L5
```

Complete the dry run table (you may not need all rows):

LINE	n	n <= 3 ?	printed

Write EXACTLY what is printed here

6. In the following dry run assume that the numbers 18, 22, 21 are input when asked and that inputInt inputs an integer. after printing the String given it as an argument, then moving to a new line.

```
int age = inputInt("First?");    // L1

while(age != 21)                // L2
{
    System.out.println(age);     // L3
    age = inputInt("Next?");     // L4
}
System.out.println(age + " so stop."); // L5
```

Complete the dry run table (you may not need all rows):

LINE	age	age != 21 ?	printed

Write EXACTLY what is printed (note this means BY THE PROGRAM so not what typed by the user) here

7. In the following dry run assume that the numbers 18, 22, 21 are input when asked and that inputInt inputs an integer. after printing the String given it as an argument, then moving to a new line.

```
int age = inputInt("First?");    // L1

while(age != 21)                // L2
{
    age = inputInt("Next?");     // L3
}
System.out.println(age + " so stop."); // L4
```

Complete the dry run table (you may not need all rows):

LINE	age	age != 21 ?	printed

Write EXACTLY what is printed (note this means BY THE PROGRAM so not what typed by the user) here

8. In the following dry run assume that the numbers 18, 22, STOP are input when asked and that inputString inputs a String. after printing the String given it as an argument, then moving to a new line.

```
final String QUIT = "STOP";           // L1
String text = inputString("First?");  // L2

while(!(text.equals("STOP")))         // L3
{
    int age = Integer.parseInt(text);  // L4
    text = inputString("Next?");       // L5
}
System.out.println(text + " so stop."); // L6
```

Complete the dry run table (you may not need all rows):

LINE	QUIT	text	age	!(text.equals("STOP")) ?	printed

Write EXACTLY what is printed (note this means BY THE PROGRAM so not what typed by the user) here